

LoRa Water Meter Security Analysis

UnicornTeam, 360 Security Technology

Who we are

The 360 Security Research Innovation Alliance consists of three teams. Each team has raised many brilliant researchers in the corresponding key areas.

UnicornTeam focuses on wireless security, and they assess the security of anything using radio technologies, from small items such as RFID/NFC, wireless sensors, drones, to large systems such as GPS, cellular networks, satellite communication systems.

RocTeam focuses on hardware development for defense and attack. They have built many hardware security products.

PegasusTeam focuses on WiFi security area. They designed 'SkyScan', a commercial wireless intrusion prevention system.

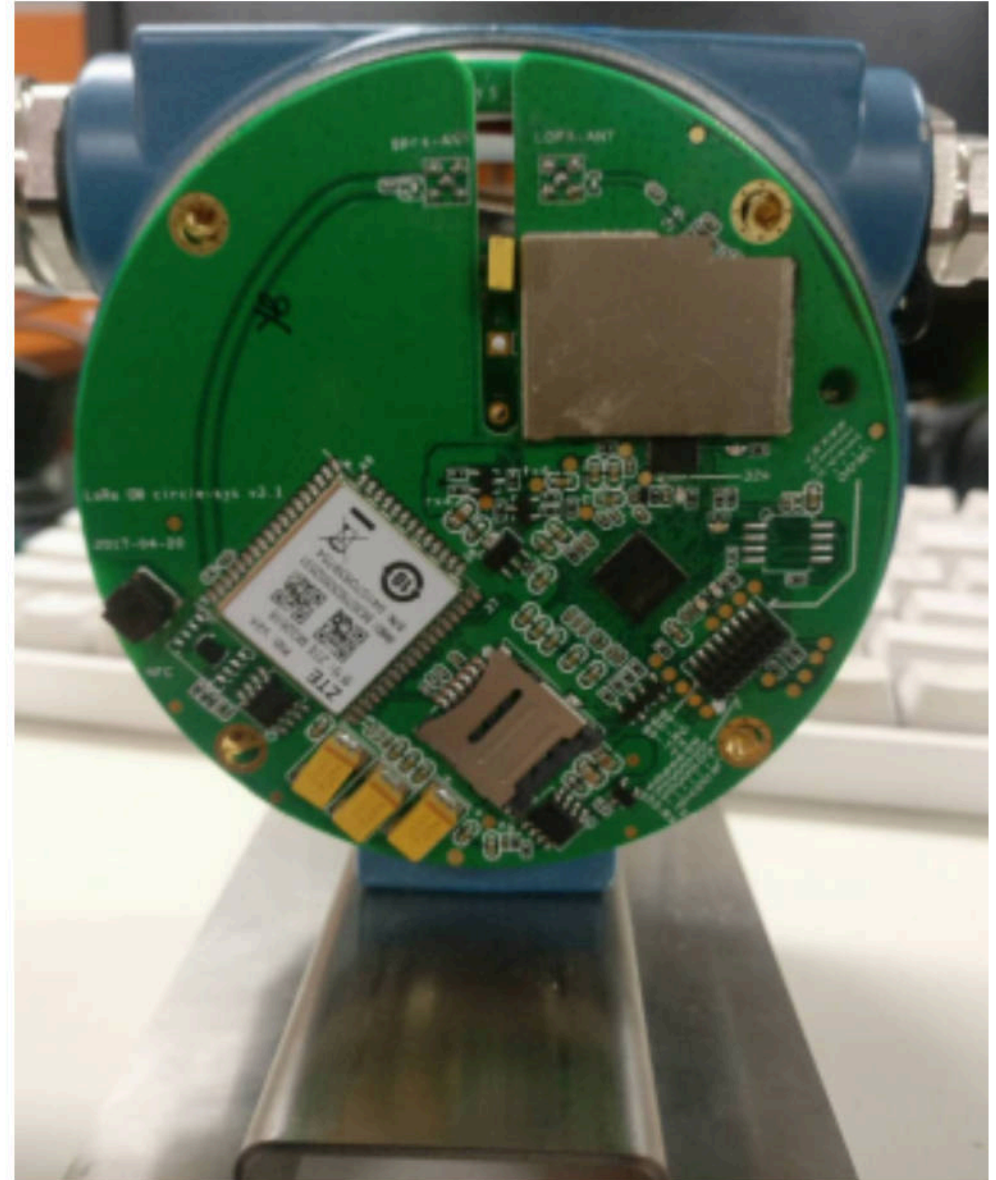


Unicorn.360.com/en

Real life Scenarios

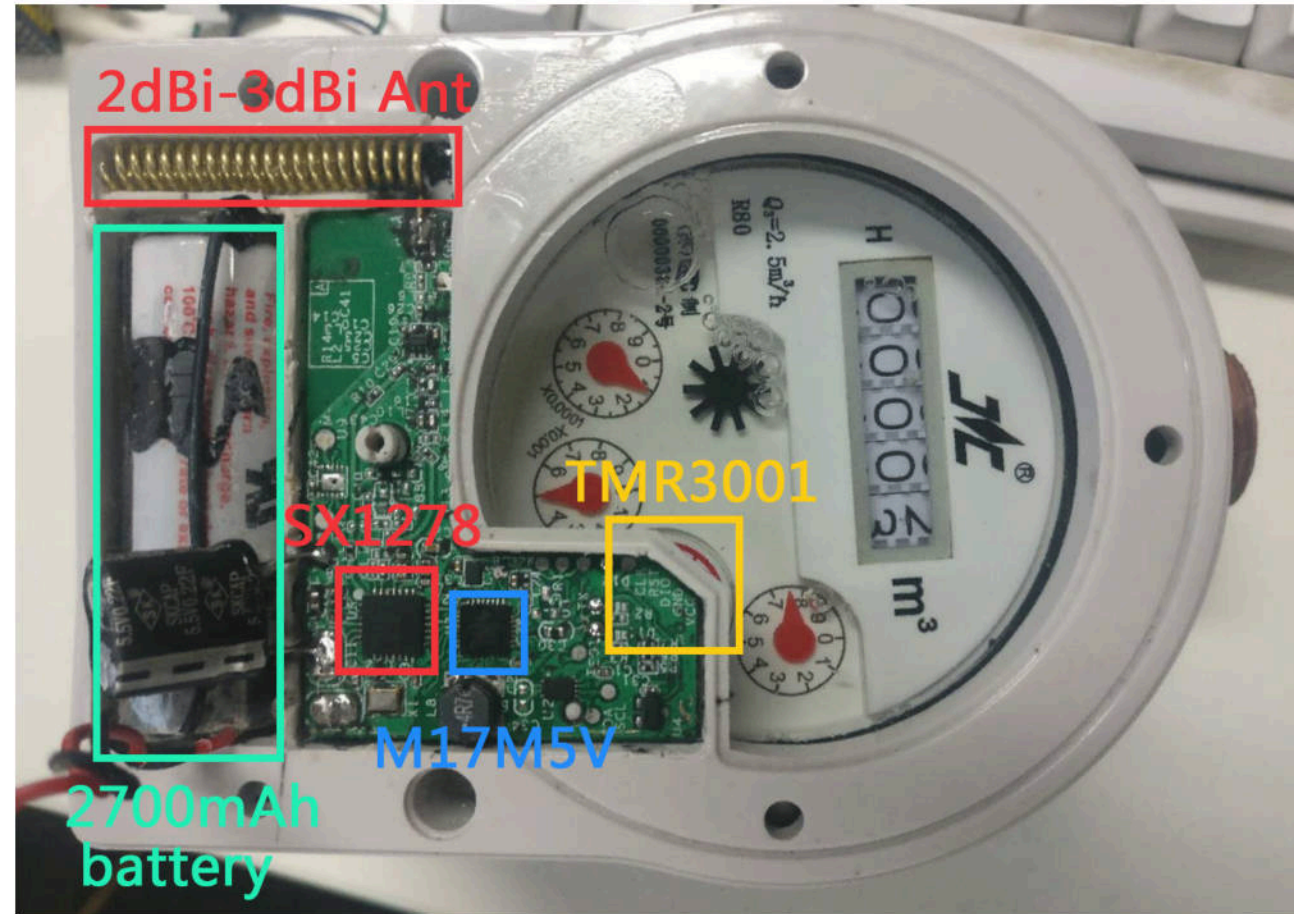


What does it look like?



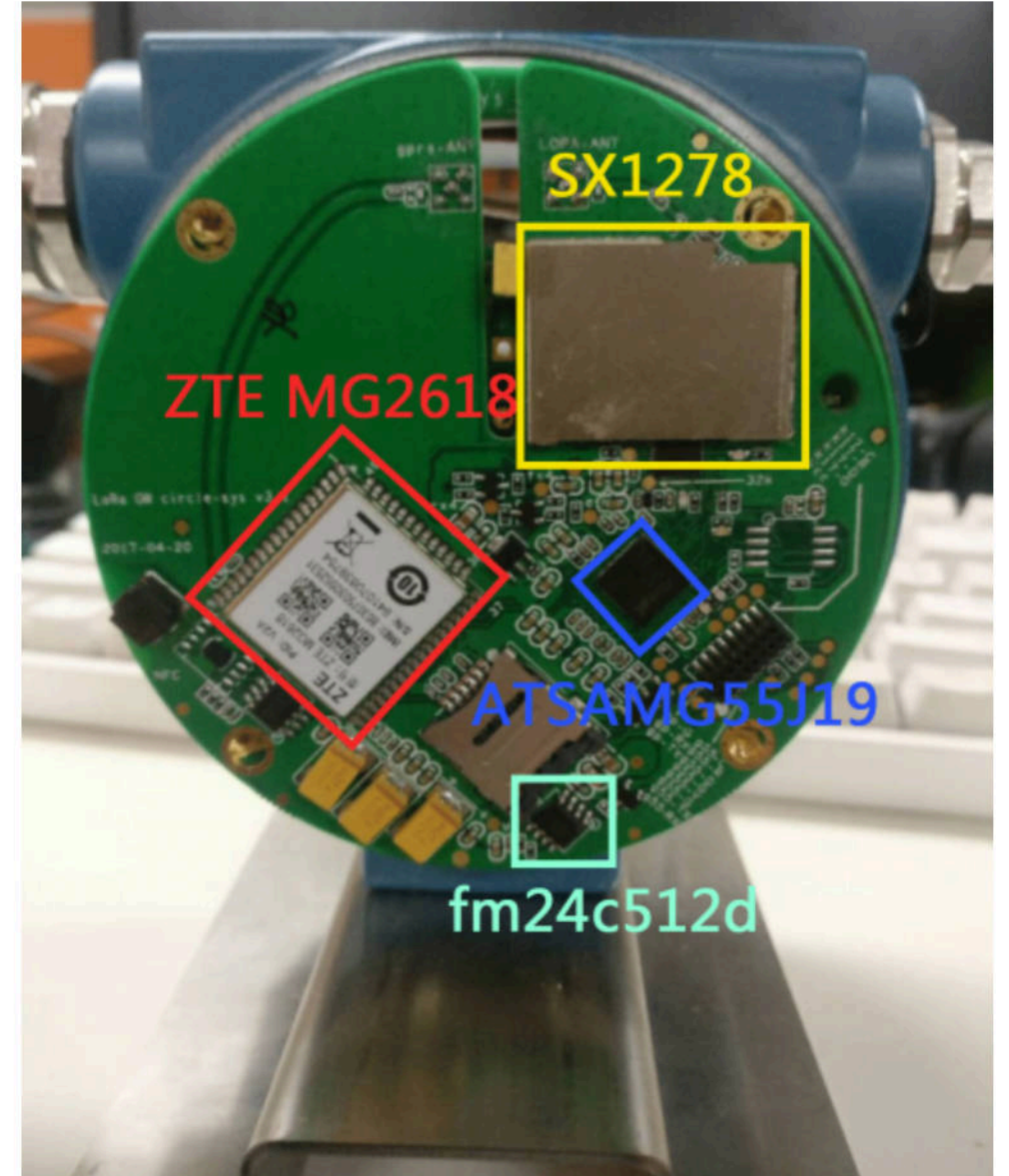
Water meter hardware disassembly

- TMR3001(Magnet sensor)
- M17M5V (NXP MCU)
- SX1278 (LORA)
- 2dBi-3dBi (ANT)
- 2700mAh (Battery)



Gateway hardware disassembly

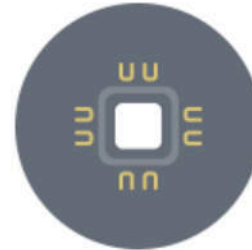
- ZTE MG2618 (GPRS)
- ATSAMG55J19 (MCU)
- SX1278 (LORA)
- FM24c512D (EEPROM)
- RT9048



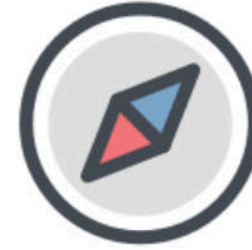
How does a water meter work?



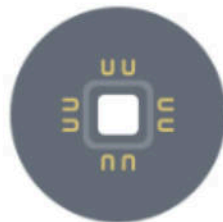
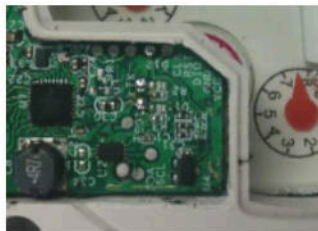
Magnet



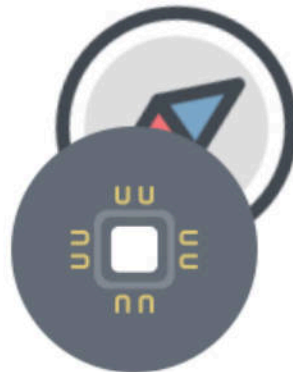
TMR3001
Magnet sensor



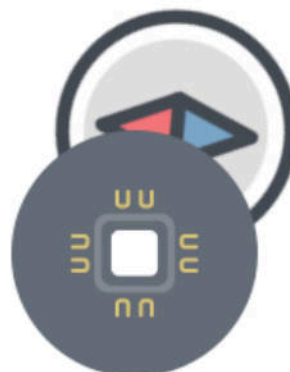
Magnet



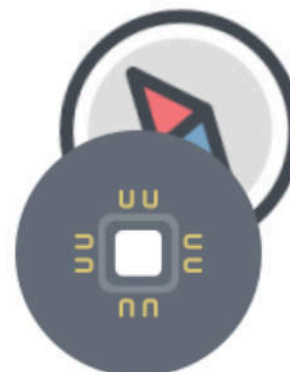
Sensor



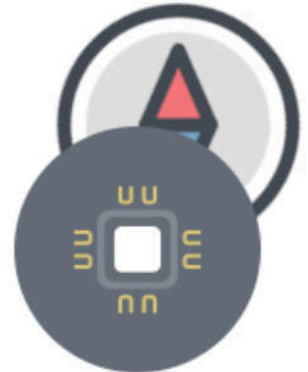
Magnet Voltage:
400 mv



Magnet Voltage:
20mv

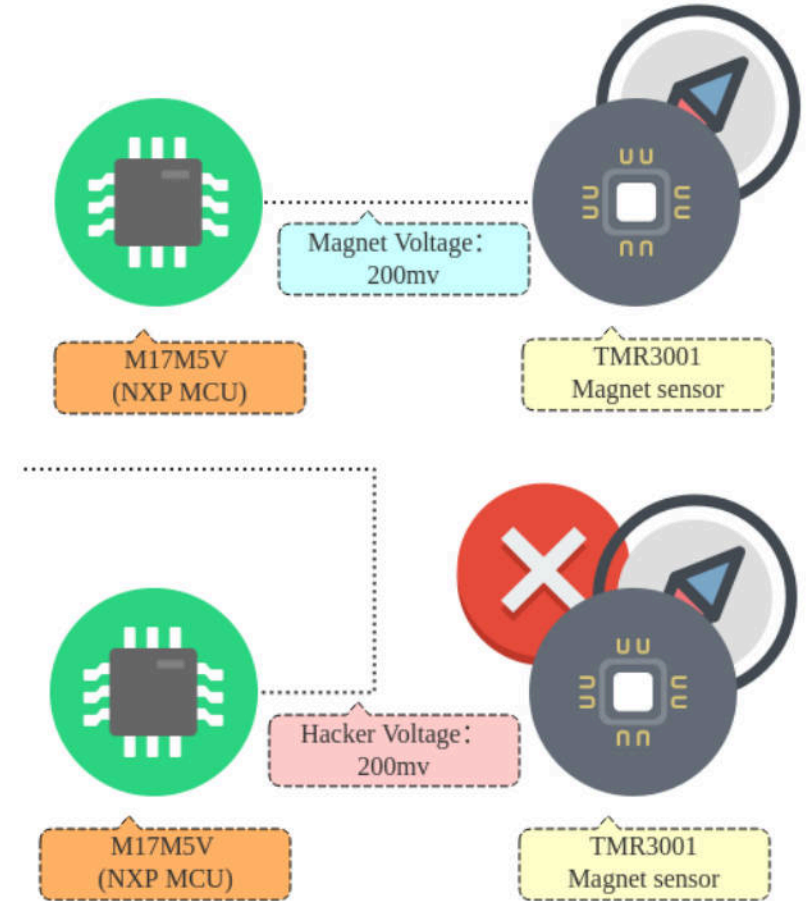
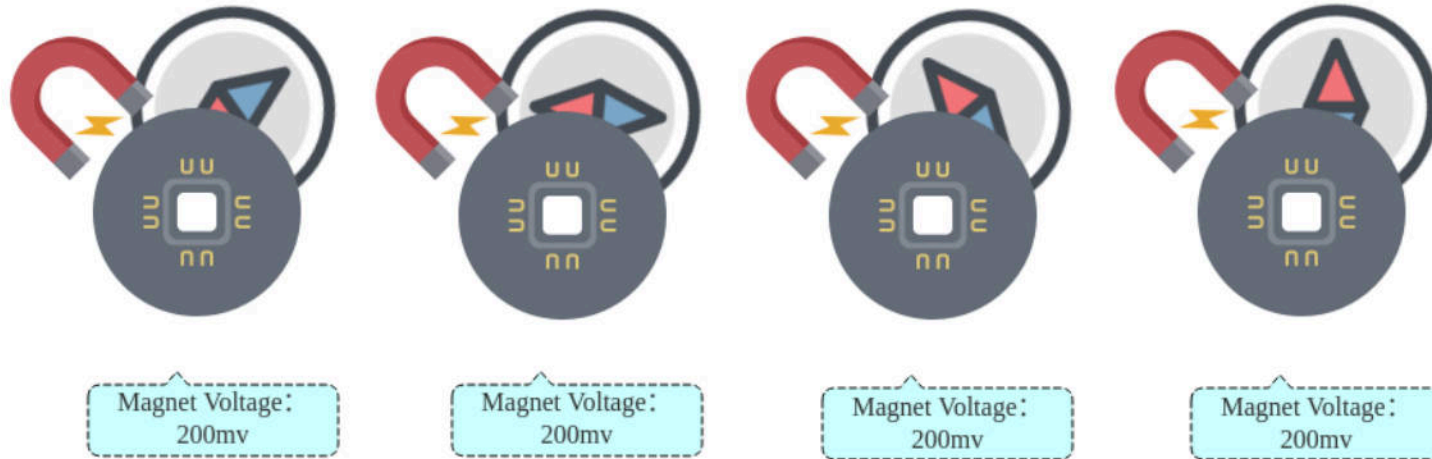
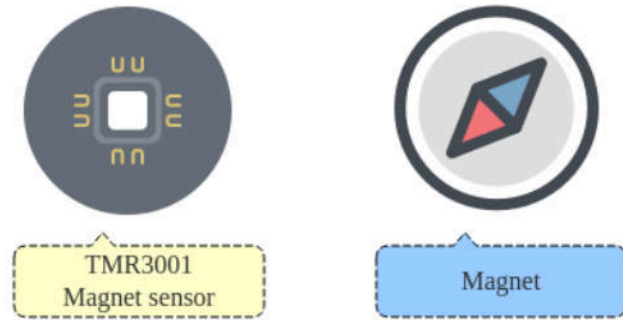


Magnet Voltage:
-200 mv

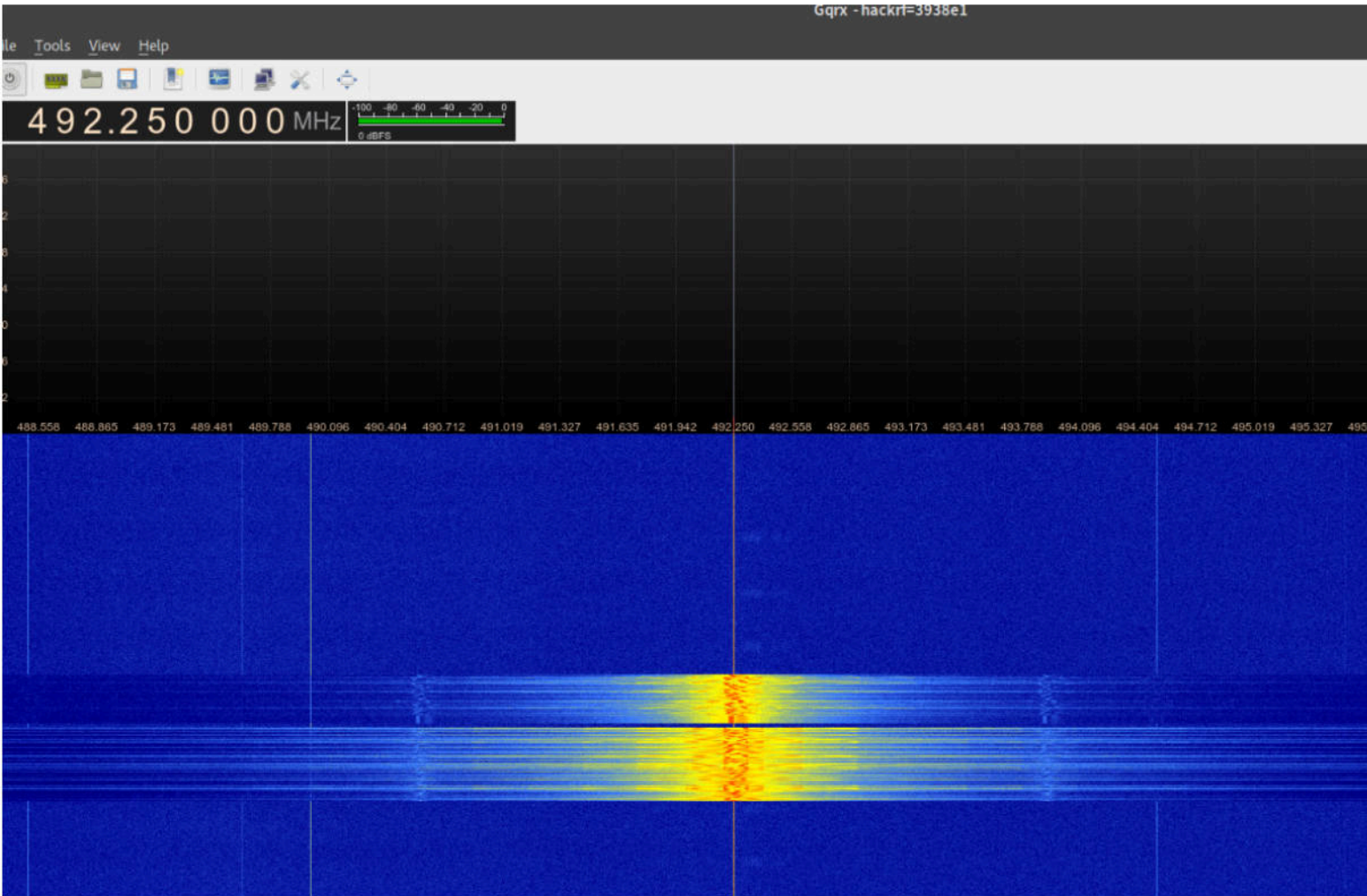


Magnet Voltage:
-300 mv

Attack the sensor

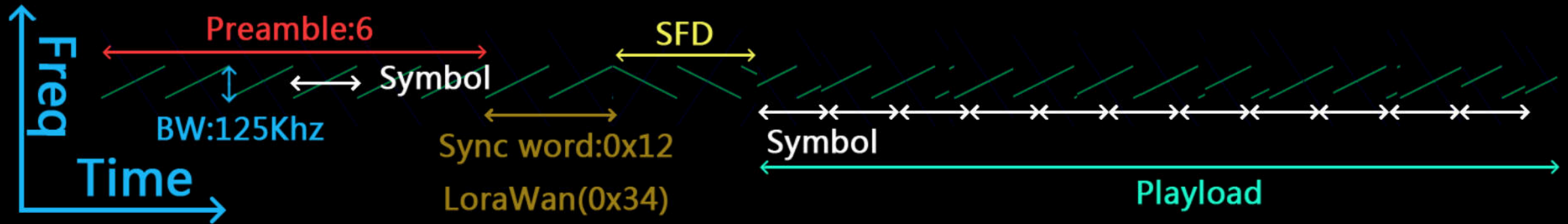


LORA Frequency



Countries	Frequency band review	Max. output power
EU	868 MHz	14 dBm
USA	915 MHz	20 dBm
Korea	900 MHz	14 dBm
Japan	920 MHz	
Malaysia	862 to 875 MHz	20 dBm
Philippines	868 MHz	
Vietnam	920 to 925 MHz	
India	865 to 867 MHz	
Singapore	922 MHz	
Thailand	920 to 925 MHz	
Indonesia	922 MHz	
ANZ	915 to 928 MHz	
Taiwan	920 to 925 MHz	
China	470 to 510 MHz	17 dBm

LORA Format



Bandwidth:125Khz(BW)

Preamble:6

Sync word:0x12

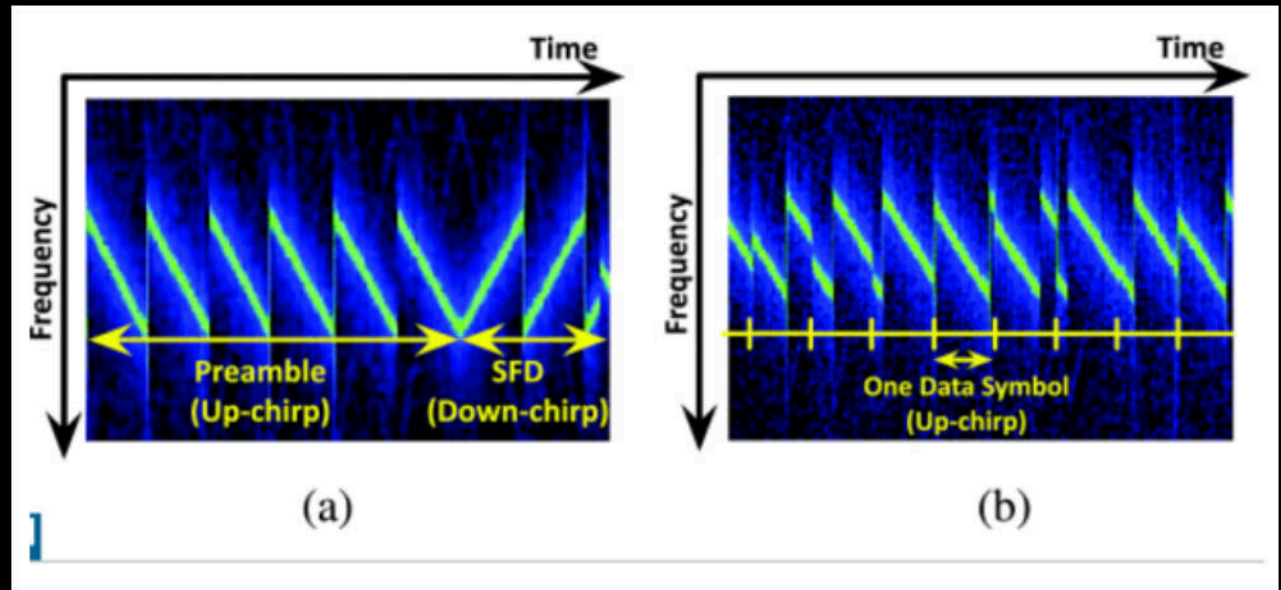
CodingRate:4/5(CR)

Spreading factor:8 (SF)

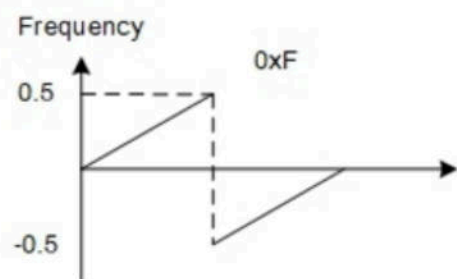
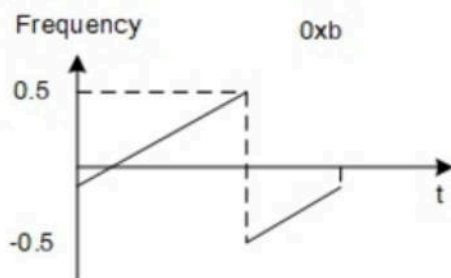
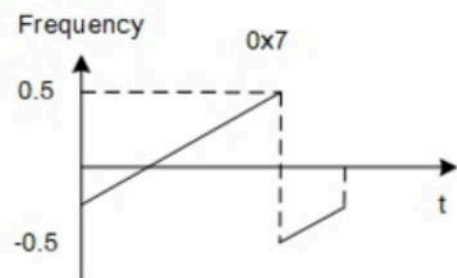
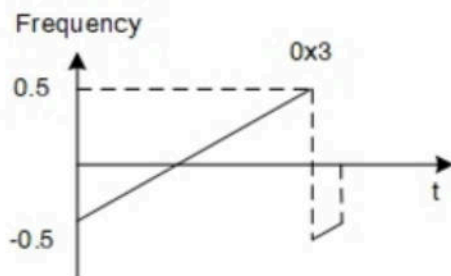
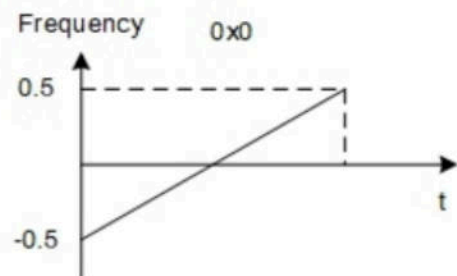
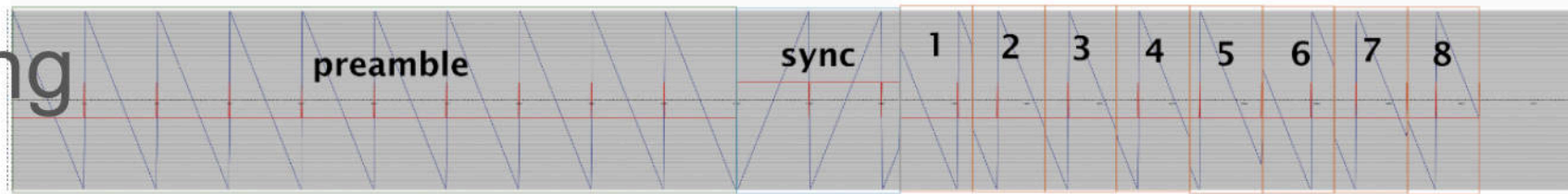
Symbol=BW/(2^SF)

DownChirp

UpChirp



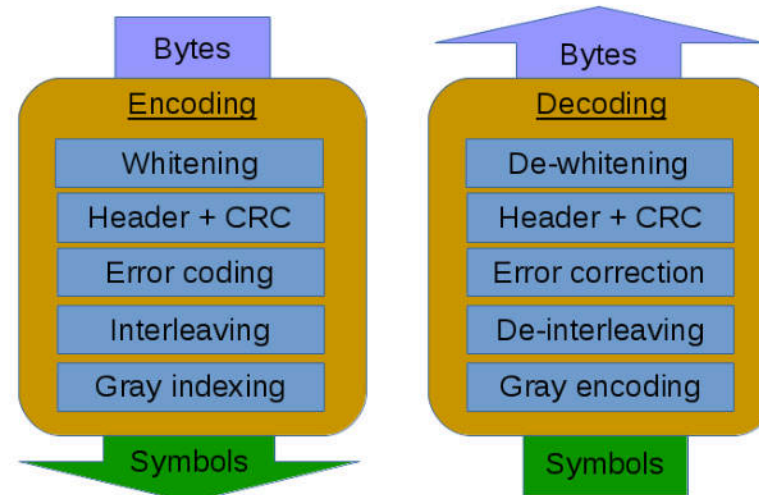
Payload encoding



<http://blog.csdn.net/ronhu>

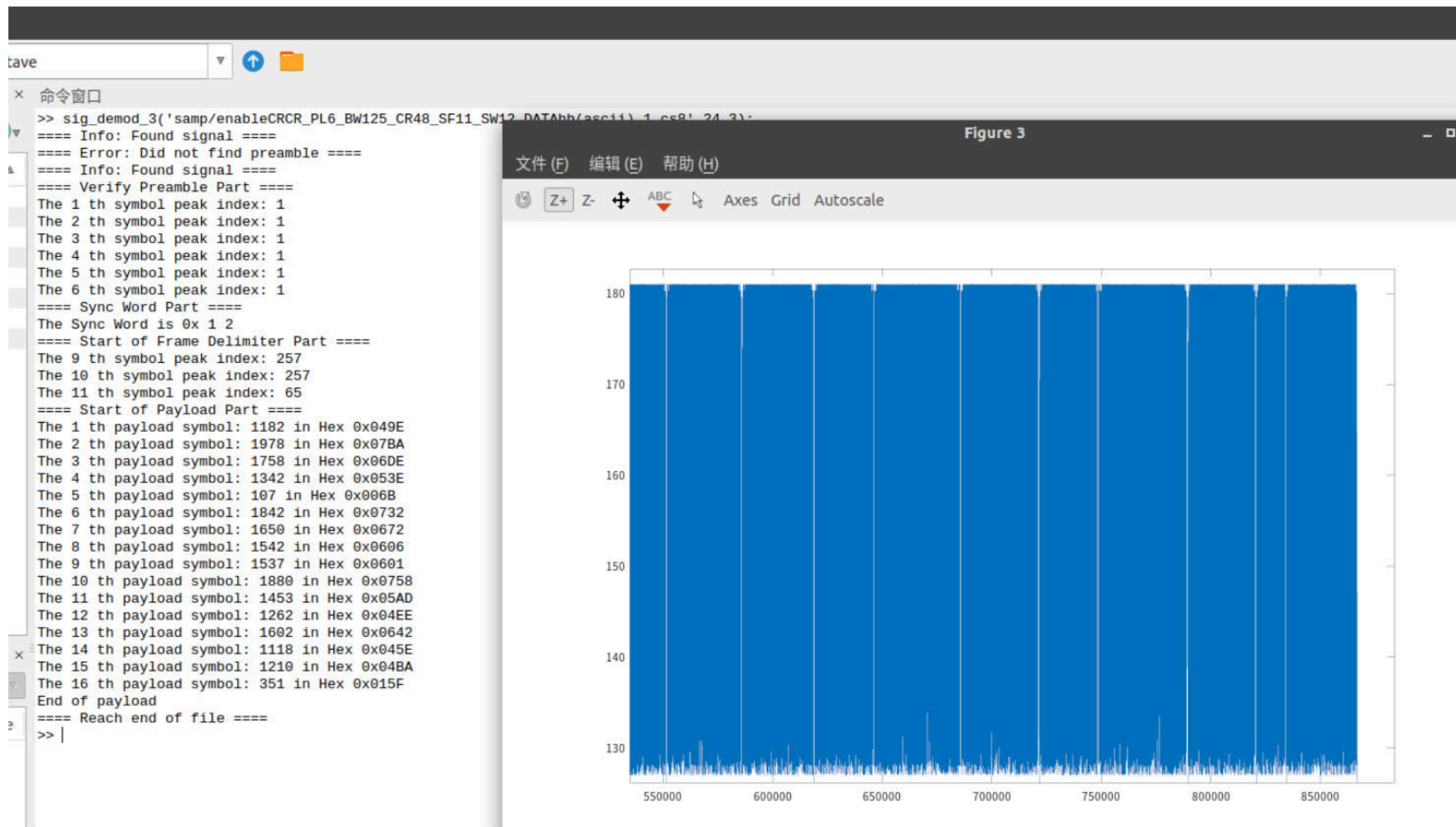
Code rate	Error Correction [bits]	Error detection [bits]
4/5	0	0
4/6	0	1
4/7	1	2
4/8	1	3

Table 2.1: Error correction and detection capabilities of LoRa



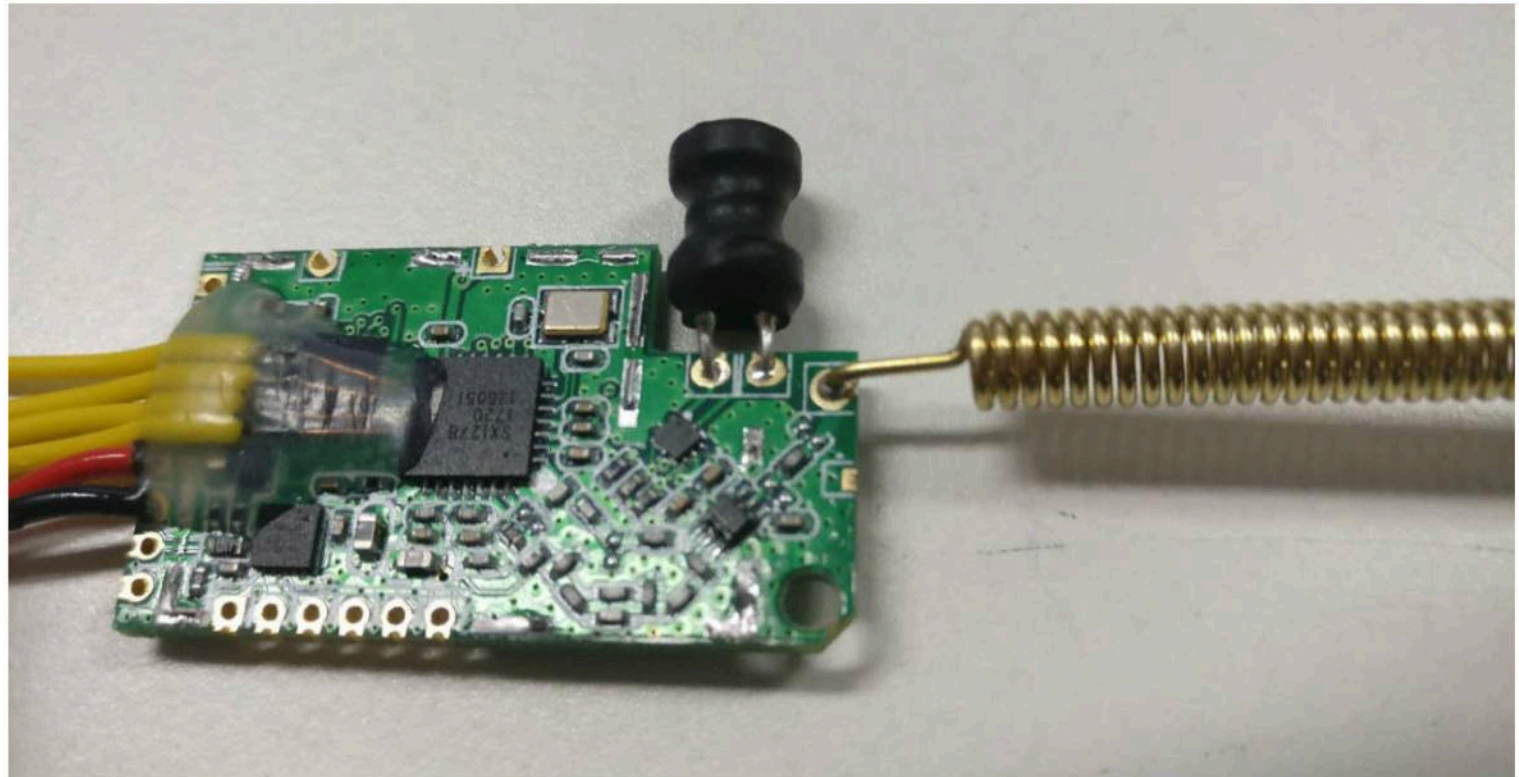
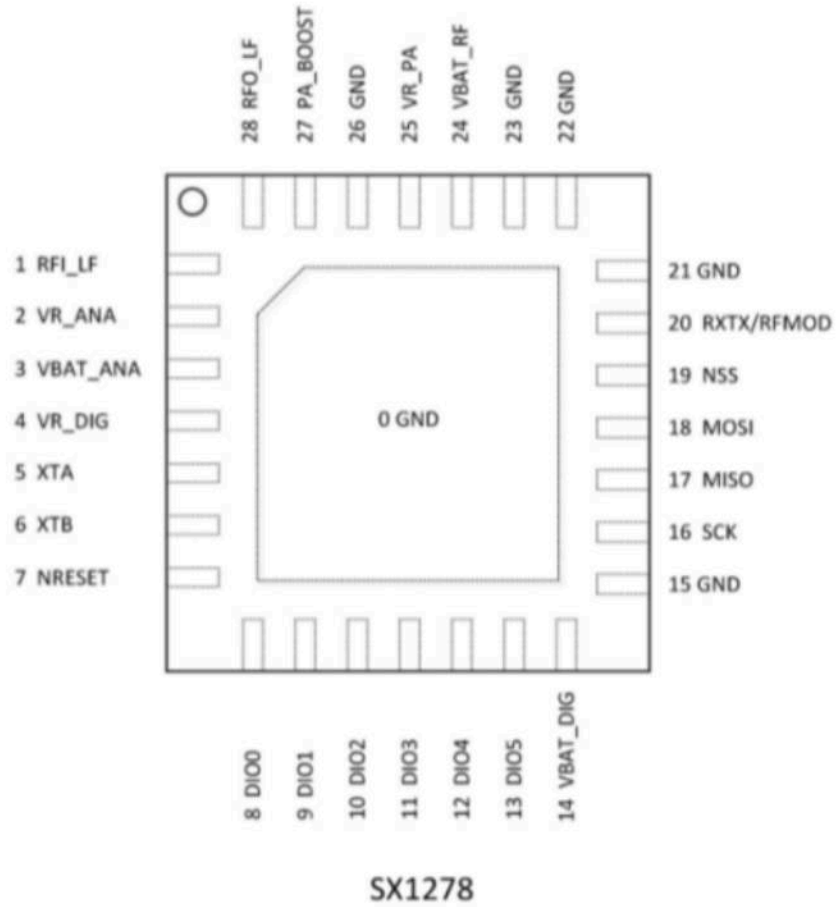
Chirp
 Whitening
 Error encoder
 Interleaver
 Gray

Demodulation lora

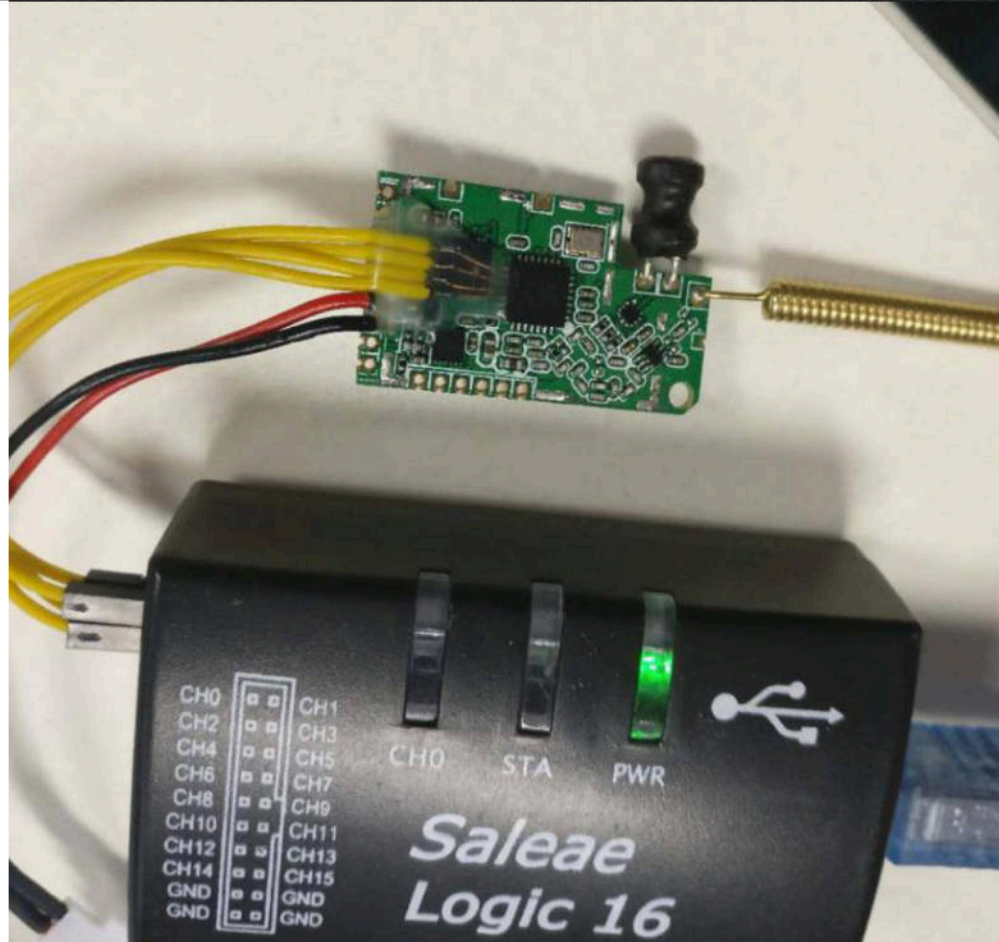
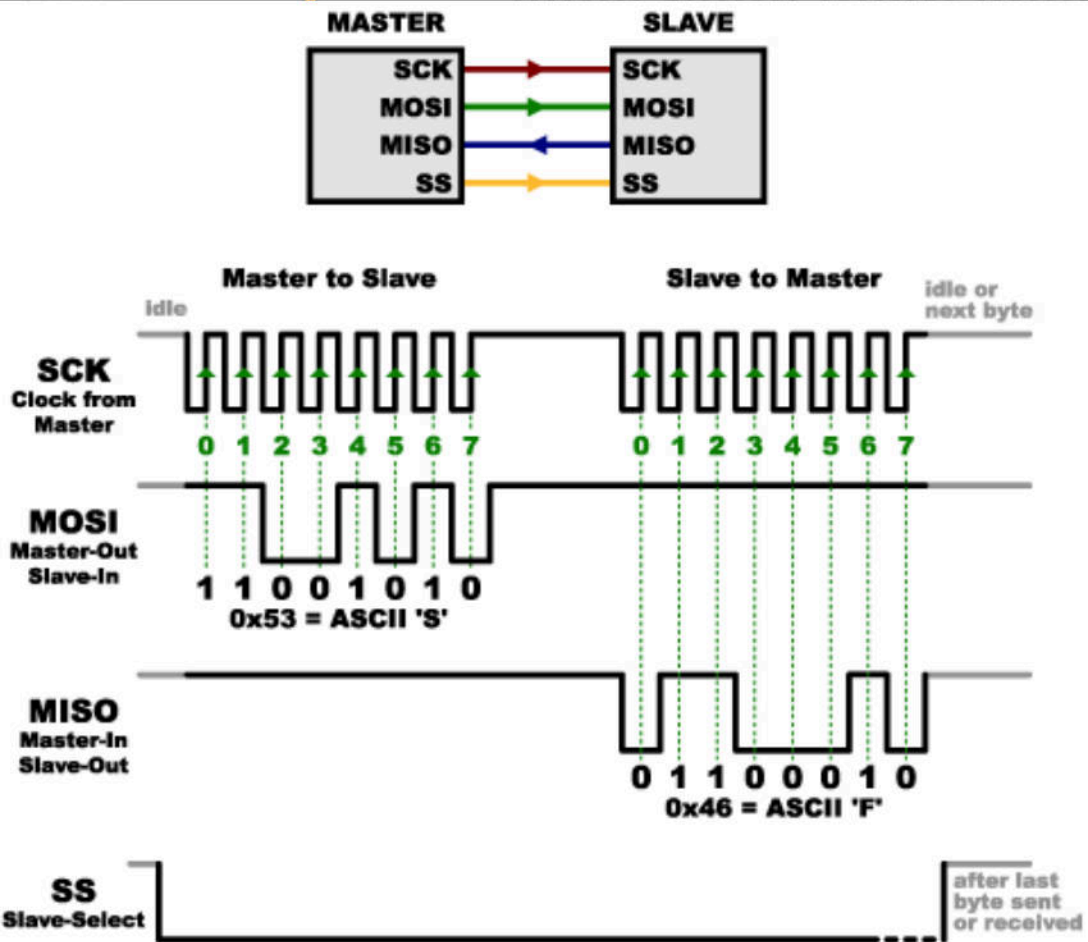
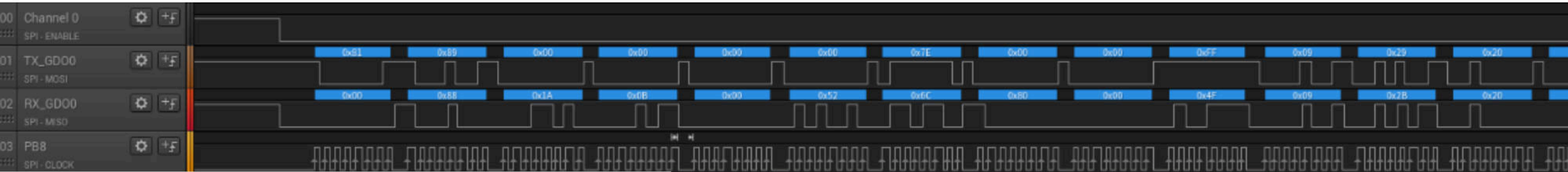


github.com/UnicornTeam/lora-demodlora

RF chip SPI sniffer



RF chip SPI sniffer



Search Protocols

MOSI: 0x92;	MISO: 0x18
MOSI: 0x10;	MISO: 0x10
MOSI: 0xC0;	MISO: 0x18
MOSI: 0x02;	MISO: 0x01
MOSI: 0x18;	MISO: 0x18
MOSI: 0x01;	MISO: 0x65
MOSI: 0x13;	MISO: 0x18
MOSI: 0x01;	MISO: 0x2F
MOSI: 0x8D;	MISO: 0x18
MOSI: 0x00;	MISO: 0x00
MOSI: 0x00;	MISO: 0x18
MOSI: 0x2F;	MISO: 0x18
MOSI: 0x06;	MISO: 0x2C
MOSI: 0x05;	MISO: 0x81
MOSI: 0x03;	MISO: 0x29
MOSI: 0x01;	MISO: 0x01
MOSI: 0x00;	MISO: 0x00
MOSI: 0x00;	MISO: 0x04
MOSI: 0x0F;	MISO: 0x81
MOSI: 0xD8;	MISO: 0x94
MOSI: 0x82;	MISO: 0x22
MOSI: 0x2E;	MISO: 0x8E
MOSI: 0x96;	MISO: 0x56
MOSI: 0x40;	MISO: 0x80

RF chip instructions reverse

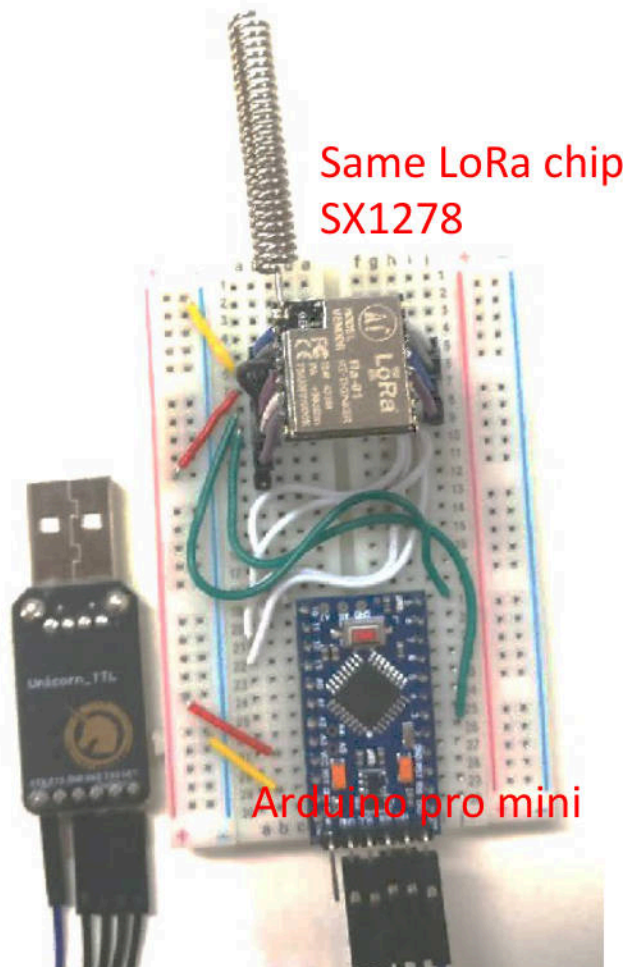
MOSI: 0x86; MISO: 0x68
 MOSI: 0x7B; MISO: 0x6C
 MOSI: 0x87; MISO: 0x68
 MOSI: 0x0F; MISO: 0x80
 MOSI: 0x88; MISO: 0x68
 MOSI: 0xFF; MISO: 0x00
 ...
 MOSI: 0x81; MISO: 0x68
 MOSI: 0x81; MISO: 0x80
 MOSI: 0xB1; MISO: 0x68
 MOSI: 0xC3; MISO: 0xC3
 MOSI: 0xB7; MISO: 0x68
 MOSI: 0x0A; MISO: 0x0A
 MOSI: 0x1E; MISO: 0x68
 MOSI: 0x00; MISO: 0x70
 MOSI: 0x9E; MISO: 0x68
 MOSI: 0x70; MISO: 0x70
 MOSI: 0x1D; MISO: 0x68
 MOSI: 0x00; MISO: 0x72
 MOSI: 0x9D; MISO: 0x68
 MOSI: 0x72; MISO: 0x72

Address	Register Name		Reset (POR)	Default (FSK)	Description	
	FSK/OOK Mode	LoRa™ Mode			FSK Mode	LoRa™ Mode
0x00	RegFifo		0x00		FIFO read/write access	
0x01	RegOpMode		0x01		Operating mode & LoRa™ / FSK selection	
0x02	RegBitrateMsb	Unused	0x1A		Bit Rate setting, Most Significant Bits	
0x03	RegBitrateLsb		0x0B		Bit Rate setting, Least Significant Bits	
0x04	RegFdevMsb		0x00		Frequency Deviation setting, Most Significant Bits	
0x05	RegFdevLsb		0x52		Frequency Deviation setting, Least Significant Bits	
0x06	RegFrFmsb		0x6C		RF Carrier Frequency, Most Significant Bits	
0x07	RegFrFmid		0x80		RF Carrier Frequency, Intermediate Bits	
0x08	RegFrFlsb		0x00		RF Carrier Frequency, Least Significant Bits	
0x09	RegPaConfig		0x4F		PA selection and Output Power control	
0x0A	RegPaRamp		0x09		Control of PA ramp time, low phase noise PLL	
0x0B	RegOcp		0x2B		Over Current Protection control	
0x0C	RegLna		0x20		LNA settings	
0x0D	RegRxConfig	RegFifoAddrPtr	0x08	0x0E	AFC, AGC, ctrl	FIFO SPI pointer
0x0E	RegRssiConfig	RegFifoTxBaseAddr	0x02		RSSI	Start Tx data
0x0F	RegRssiCollision	RegFifoRxBaseAddr	0x0A		RSSI Collision detector	Start Rx data
0x10	RegRssiThresh	FifoRxCurrentAddr	0xFF		RSSI Threshold control	Start address of last packet received
0x11	RegRssiValue	RegIrqFlagsMask	n/a	n/a	RSSI value in dBm	Optional IRQ flag mask
0x12	RegRxBw	RegIrqFlags	0x15		Channel Filter BW Control	IRQ flags
0x13	RegAfcBw	RegRxBnBytes	0x0B		AFC Channel Filter BW	Number of received bytes
0x14	RegOokPeak	RegRxHeaderCntValueMsb	0x28		OOK demodulator	Number of valid headers received
0x15	RegOokFix	RegRxHeaderCntValueLsb	0x0C		Threshold of the OOK demod	
0x16	RegOokAvg	RegRxPacketCntValueMsb	0x12		Average of the OOK demod	Number of valid packets received
0x17	Reserved17	RegRxPacketCntValueLsb	0x47		-	
0x18	Reserved18	RegModemStat	0x32		-	Live LoRa™ modem status
0x19	Reserved19	RegPktSnrValue	0x3E		-	Estimation of last packet SNR
0x1A	RegAfcFei	RegPktRssiValue	0x00		AFC and FEI control	RSSI of last packet

RF chip instructions reverse

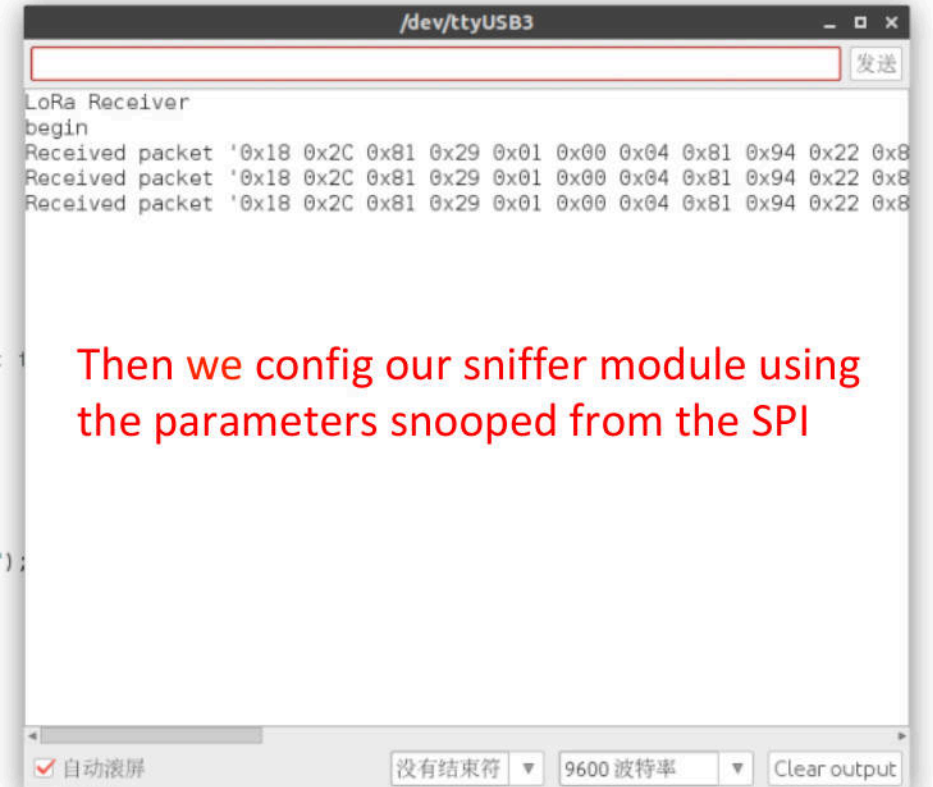
```
MOSI: 0x42; MISO: 0x00
MOSI: 0x00; MISO: 0x12    //Read VERSION
MOSI: 0x86; MISO: 0x68
MOSI: 0x7B; MISO: 0x6C
MOSI: 0x87; MISO: 0x68
MOSI: 0x0F; MISO: 0x80
MOSI: 0x88; MISO: 0x68
MOSI: 0xFF; MISO: 0x00    // Set frequency 492.25 Mhz
MOSI: 0xA6; MISO: 0x68
MOSI: 0x04; MISO: 0x04    // Set the AGC LNA gain
MOSI: 0x81; MISO: 0x68
MOSI: 0x81; MISO: 0x80    // Enter IDLE
MOSI: 0xB1; MISO: 0x68    // writeRegister(REG_DETECTION_OPTIMIZE, 0xc3);
MOSI: 0xC3; MISO: 0xC3    // Set SF
MOSI: 0xB7; MISO: 0x68    // writeRegister(REG_DETECTION_THRESHOLD, 0x0a);
MOSI: 0x0A; MISO: 0x0A    // Set SF
MOSI: 0x1E; MISO: 0x68    // readRegister(REG_MODEM_CONFIG_2) & 0x0f)
MOSI: 0x00; MISO: 0x70    // Set SF
MOSI: 0x9E; MISO: 0x68    // writeReg(readRegister(REG_MODEM_CONFIG_2) & 0x0f) | ((sf << 4) & 0xf0));
MOSI: 0x70; MISO: 0x70    // Set SF 7
MOSI: 0x1D; MISO: 0x68    // writeReg (readRegister(REG_MODEM_CONFIG_1) & 0x0f)
MOSI: 0x00; MISO: 0x72    // Set BW CR Headr
MOSI: 0x9D; MISO: 0x68    // writeReg (readRegister(REG_MODEM_CONFIG_1) & 0x0f) | (bw << 4));
MOSI: 0x72; MISO: 0x72    // Set BW 125Khz CR4/5 Explicit Header mode
```


Lora Sniffer hardware



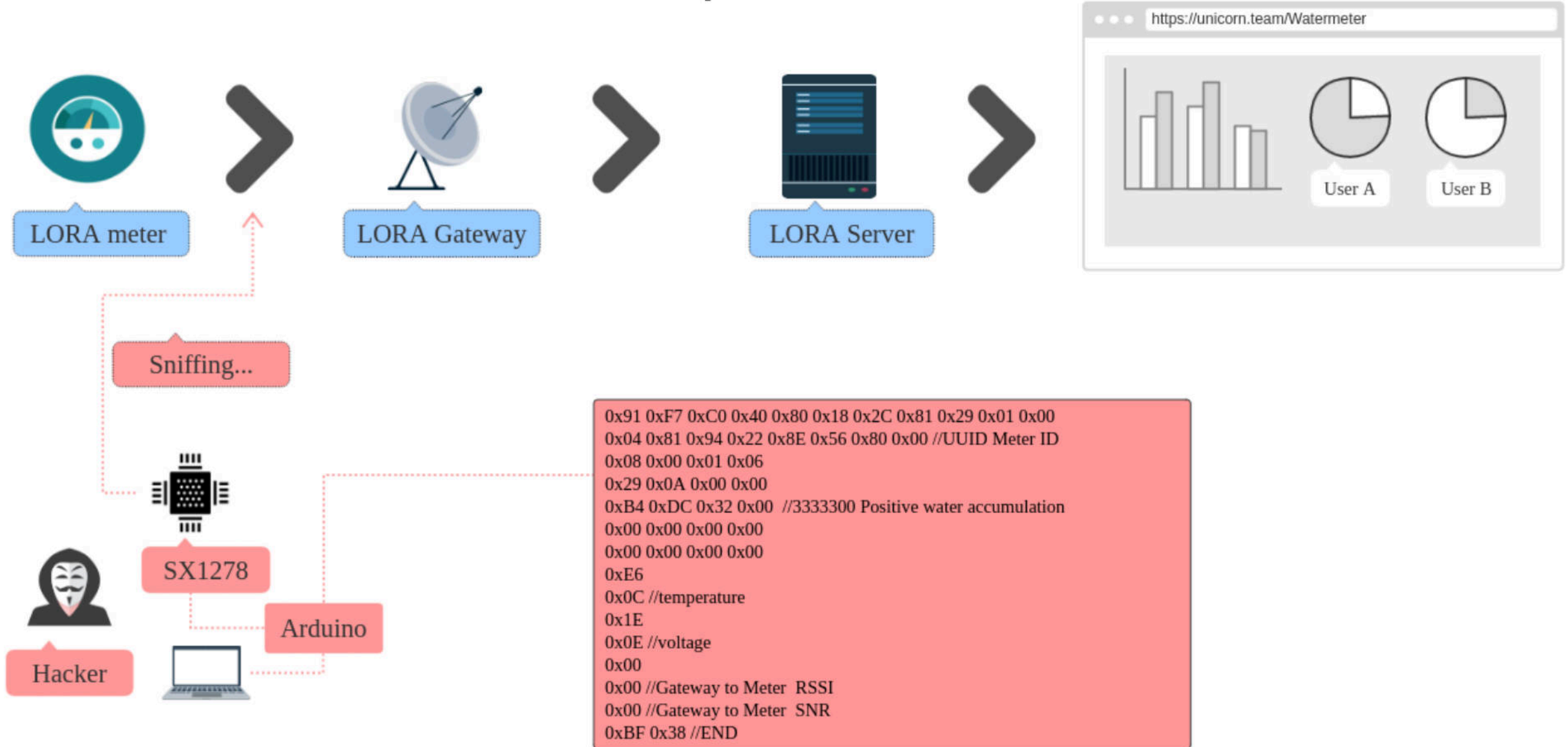
```
Sniffer_Watermeter_ESP32_simulation
//SPI.transfer(0x00);
//SPI.transfer(0x00);
//SPI.transfer(0x00);
//SPI.transfer(0x00);
//SPI.transfer(0x7E);
//SPI.transfer(0x00);
//SPI.transfer(0x00);
//SPI.transfer(0xFF);
//SPI.transfer(0x09);
//SPI.transfer(0x29);
//SPI.transfer(0x20);
//LoRa.writeRegister(0x01, 0x88);
//LoRa.writeRegister(0x01, 0x88);
}
void setup() {
  Serial.begin(9600);
  while (!Serial); //if just the the basic
  delay(1000);

  Serial.println("LoRa Receiver");
  //Simulation SPI();
  SPI.begin(SCK, MISO, MOSI, SS);
  LoRa.setPins(SS, RST, DI00);
  if (!LoRa.begin(BAND, PAB00ST)) {
    Serial.println("Starting LoRa failed!");
    while (1);
  }
  LoRa.implicitHeaderMode();
  //LoRa.explicitHeaderMode();
  //LoRa.explicitHeaderMode();
  LoRa.setSpreadingFactor(11);
  LoRa.setCodingRate4(5);
  LoRa.setSignalBandwidth(125E3);
  LoRa.setPreambleLength(7);
  LoRa.setSyncWord(0x12);
  //LoRa.disableCrc();
  LoRa.crc();
  Serial.println("begin");
  LoRa.receive();
}
```

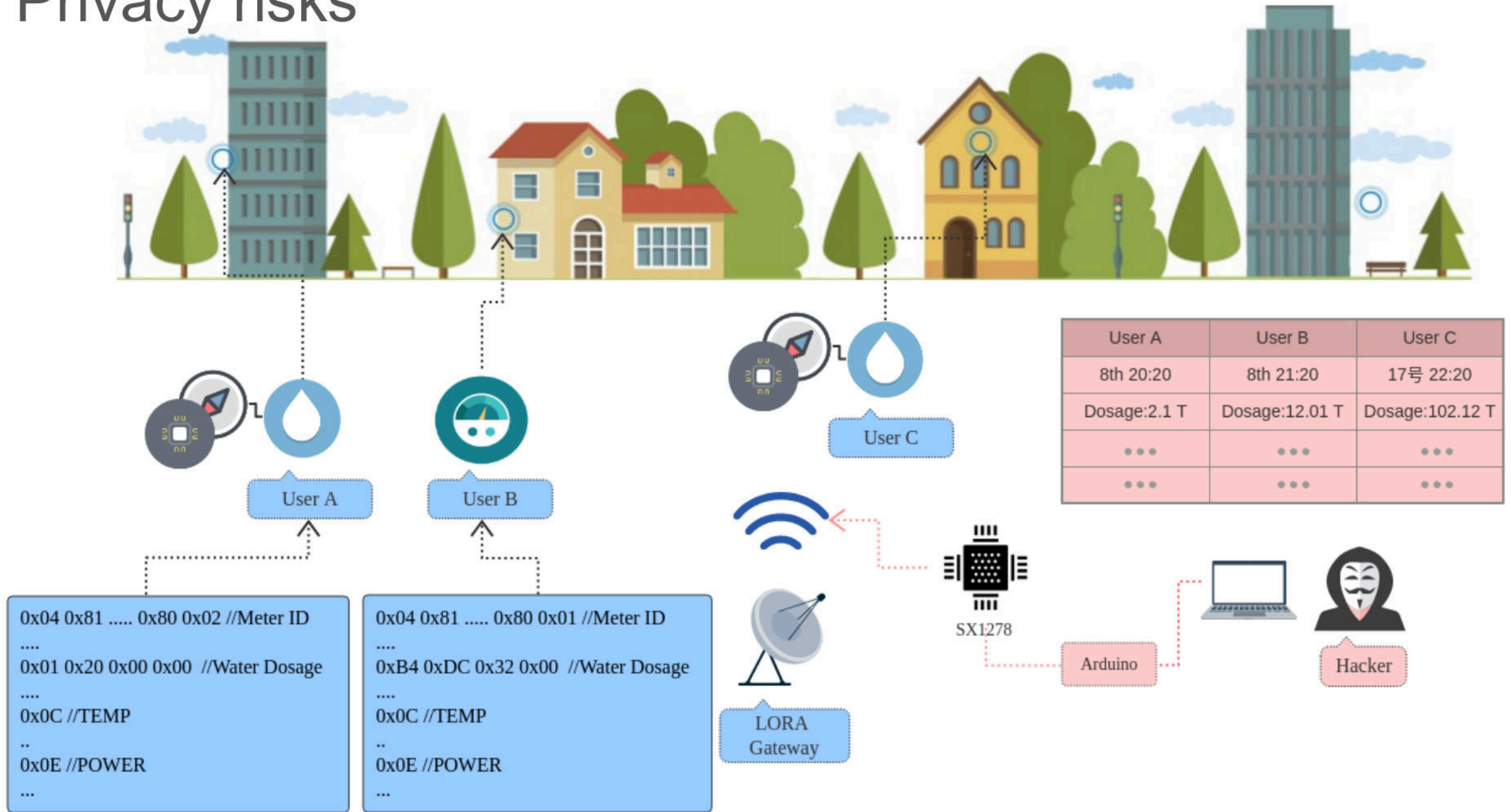


Then we config our sniffer module using the parameters snooped from the SPI

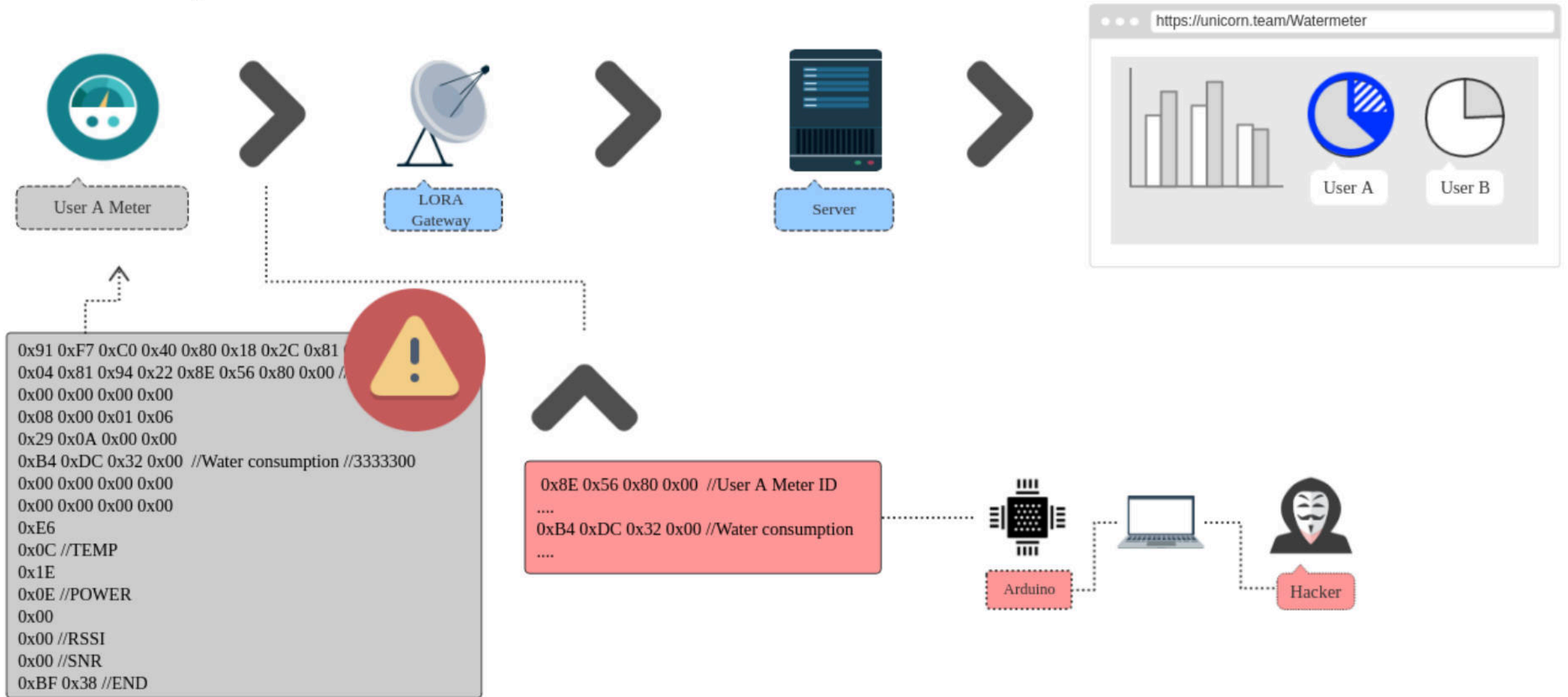
Reverse communication protocol



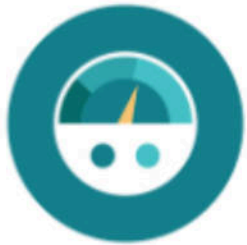
Privacy risks



Fake upload data



Forge the gateway to issue instructions.



UserA LORA Meter

Meter ID: 0x8E 0x56 0x80 0x00



Gas Valve

Valve ID: 0x8E 0x56 0x80 0x01

Use sx1278 to enter sniff mode

....

Sniffer LORA Network ID info

....

//Sniffing the ID and then forging the gateway to transmit malicious instructions

....

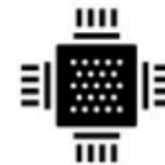
Close ID 0x00 Valve

....

Close ID 0x01 Valve



Forged
LORA Gateway

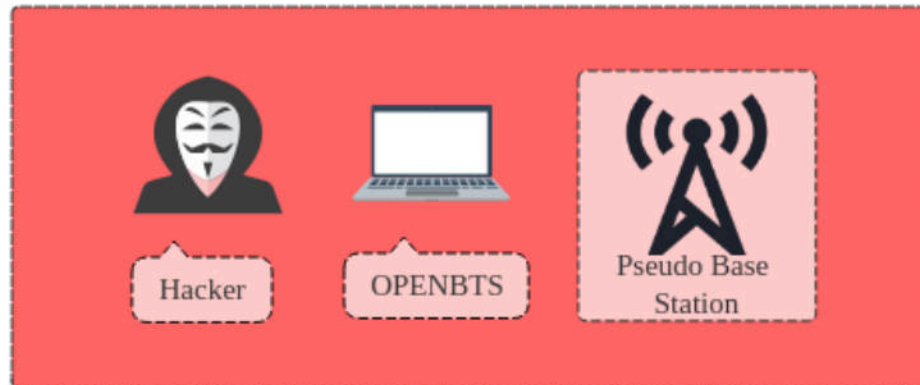
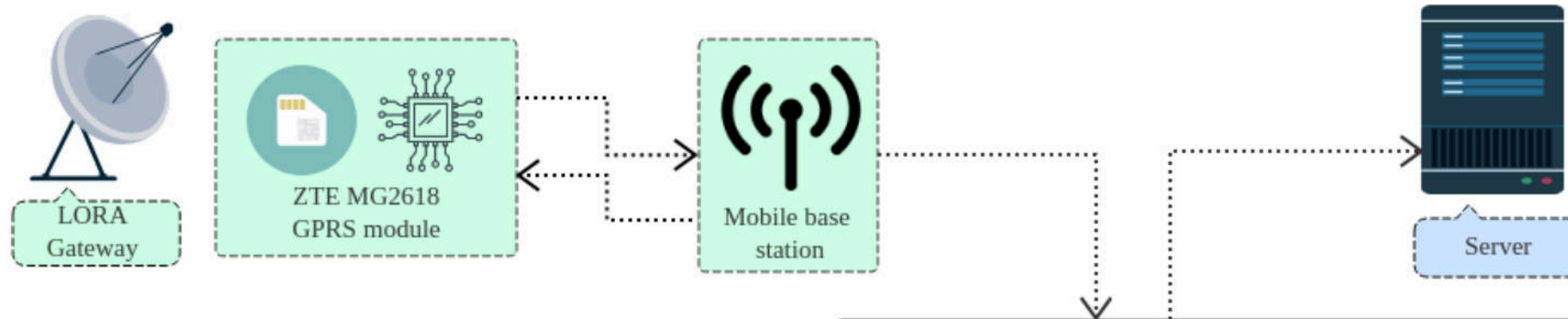


SX1278



Hacker

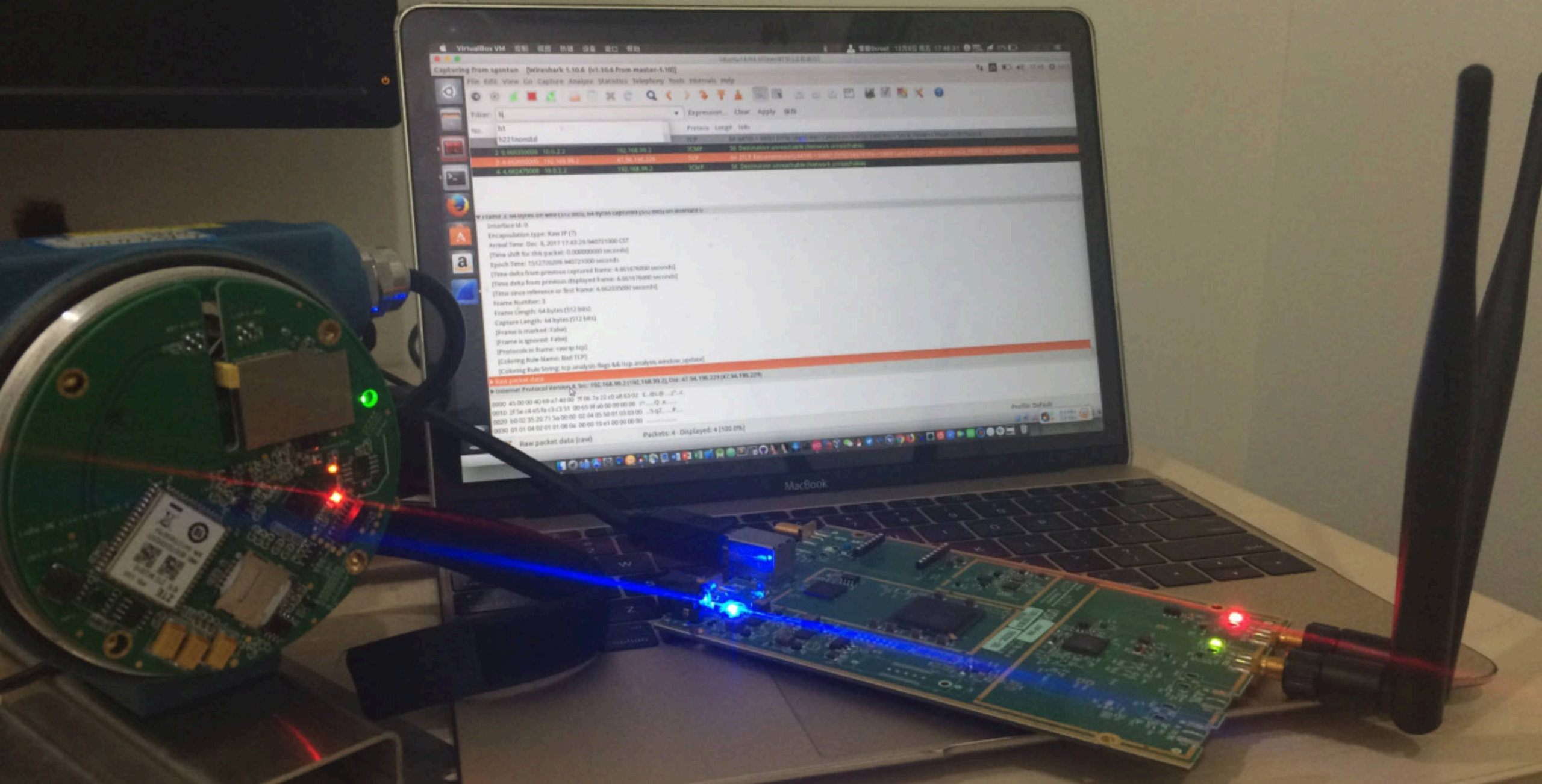
GPRS communication link



```
AA 00 28 7A 2C 0B 6E 81 59 02 E0 AA 42 00 1D 00 22 04 81 94 22 8E 56 80 00 00 32 DC B4 00
00 00 00 00 00 00 00 00 0E 0C E6 1E 17 01 19 11 25 65 0B 00 00 84 55

7A 2C 0B 6E 81 59 02 E0 //Gateway ID
AA //Header
42 //Cont
??00
1D //application
00 22 //Len
04 81 94 22 8E 56 80 00 //UUID Meter ID
00 32 DC B4 //3333300 Positive water accumulation
00 00 00 00/ //Negative cumulative water volume
00 00
00 00 //Meter State
00 0E //Voltage
0C //temperature
?: E6 1E
17 01 19 11 25 //Meter reading time
65 //Meter to Gateway RSSI
0b //Meter to Gateway SNR
00 //Gateway to Meter RSSI
00 //Gateway to Meter SNR
84 //CRC
```

GPRS Man-in-the-MiddleAttack



00000000

Data (data) 19 bytes Packets: 33, Displayed: Profile: Default

No.	Time	Source	Destination	Protocol	Length	Info
3	295.181109		.2	TCP	64	50614 → 50001 [SYN] Seq=0 Win=10880 Len=0 MSS=1360 WS=1 SACK_PERM=1 TSval=22915 TSecr=0
4	295.187401		229	TCP	44	50001 → 50614 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460
5	295.479558		.2	TCP	40	50614 → 50001 [ACK] Seq=1 Ack=1 Win=10880 Len=0
6	295.538890		.2	TCP	65	50614 → 50001 [PSH, ACK] Seq=1 Ack=1 Win=10880 Len=25
7	295.539131		229	TCP	40	50001 → 50614 [ACK] Seq=1 Ack=26 Win=65535 Len=0
9	295.544835		229	TCP	59	50001 → 50614 [PSH, ACK] Seq=1 Ack=26 Win=65535 Len=19
14	296.554644		.2	TCP	59	50614 → 50001 [PSH, ACK] Seq=26 Ack=20 Win=10880 Len=19
15	296.554991		229	TCP	40	50001 → 50614 [ACK] Seq=20 Ack=45 Win=65535 Len=0
16	296.560530		229	TCP	65	50001 → 50614 [PSH, ACK] Seq=20 Ack=45 Win=65535 Len=25
18	297.137181		.2	TCP	40	50614 → 50001 [ACK] Seq=45 Ack=45 Win=10880 Len=0
20	337.461784		.2	TCP	65	50614 → 50001 [PSH, ACK] Seq=45 Ack=45 Win=10880 Len=25
21	337.462168		229			
22	337.462259		.2			
23	337.462433		229			
24	337.467670		229			
25	337.518138		.2			
26	337.518370		229			
27	337.561945		.2			
28	337.563148		229			
29	338.126089		.2			
30	358.289074		.2			
31	358.289447		229			
32	358.295159		229			
33	358.771720		.2			

Wireshark · 追踪 TCP 流 (tcp.stream eq 1) · iot_fuzz

00000000 aa 00 0c 7a 2c 0b 6e 81 59 02 e0 aa 41 00 02 00 ...z,.n. Y...A...
00000010 06 6c 78 08 21 01 96 7e 55 ...lx.!...~ U

00000000 aa 00 06 7a 2c 0b 6e 81 59 02 e0 aa 80 00 00 00 ...z,.n. Y.....
00000010 00 0b 55 ..U

00000019 aa 00 06 7a 2c 0b 6e 81 59 02 e0 aa 43 00 12 00 ...z,.n. Y...C...
00000029 00 e0 55 ..U

00000013 aa 00 0c 7a 2c 0b 6e 81 59 02 e0 aa 81 00 12 00 ...z,.n. Y.....
00000023 06 17 01 19 11 24 04 94 55\$. U

0000002C aa 00 0c 7a 2c 0b 6e 81 59 02 e0 aa 41 00 11 00 ...z,.n. Y...A...
0000003C 06 00 00 00 00 00 00 e9 55 U

0000002C aa 00 0c 7a 2c 0b 6e 81 59 02 e0 aa 80 00 0b 04 ...z,.n. Y.....
0000003C 7e 94 22 8e 56 80 00 b8 55 ~."V... U

00000045 aa 00 0c 7a 2c 0b 6e 81 59 02 e0 aa 41 00 11 00 ...z,.n. Y...A...
00000055 06 00 00 00 00 00 00 e9 55 U

00000045 aa 00 0c 7a 2c 0b 6e 81 59 02 e0 aa 80 00 0b 04 ...z,.n. Y.....
00000055 7e 94 22 8e 56 80 00 b8 55 ~."V... U

6 客户端 分组, 6 服务器 分组, 9 turn(s).

Entire conversation (188 bytes) 显示和保存数据为 Hex 转储 流 1

查找: 查找下一个(N)

Help 滤掉此流 打印 Save as... 返回 Close

Window size value: 65535
[Calculated window size: 65535]
[Window size scaling factor: -2 (no
Checksum: 0xaea5 [unverified]
[Checksum Status: Unverified]
Urgent pointer: 0
[SEQ/ACK analysis]
[iRTT: 0.298449000 seconds]
[Bytes in flight: 25]
[Bytes sent since last PSH flag: 2
TCP payload (25 bytes)
Data (25 bytes)

Data: aa000c7a2c0b6e815902e0aa80000b047e94228e568000b8...

0000
0010

Reverse upload protocol

recv:AA 00 0C 7A 2C 0B 6E 81 59 02 E0 AA 41 00 02 00 06 6C 78 08 21 01 96 7E 55 //Gateway registration
Server:aa 00 06 7a 2c 0b 6e 81 59 02 e0 aa 80 00 00 00 00 0b 55 //Service returns the packet

recv:AA 00 06 7A 2C 0B 6E 81 59 02 E0 AA 43 00 12 00 00 E0 55 //Gateway request time
Server:aa 00 0c 7a 2c 0b 6e 81 59 02 e0 aa 81 00 12 00 06 17 01 19 11 24 04 94 55 //Server return time 17 01 19 11 24 04

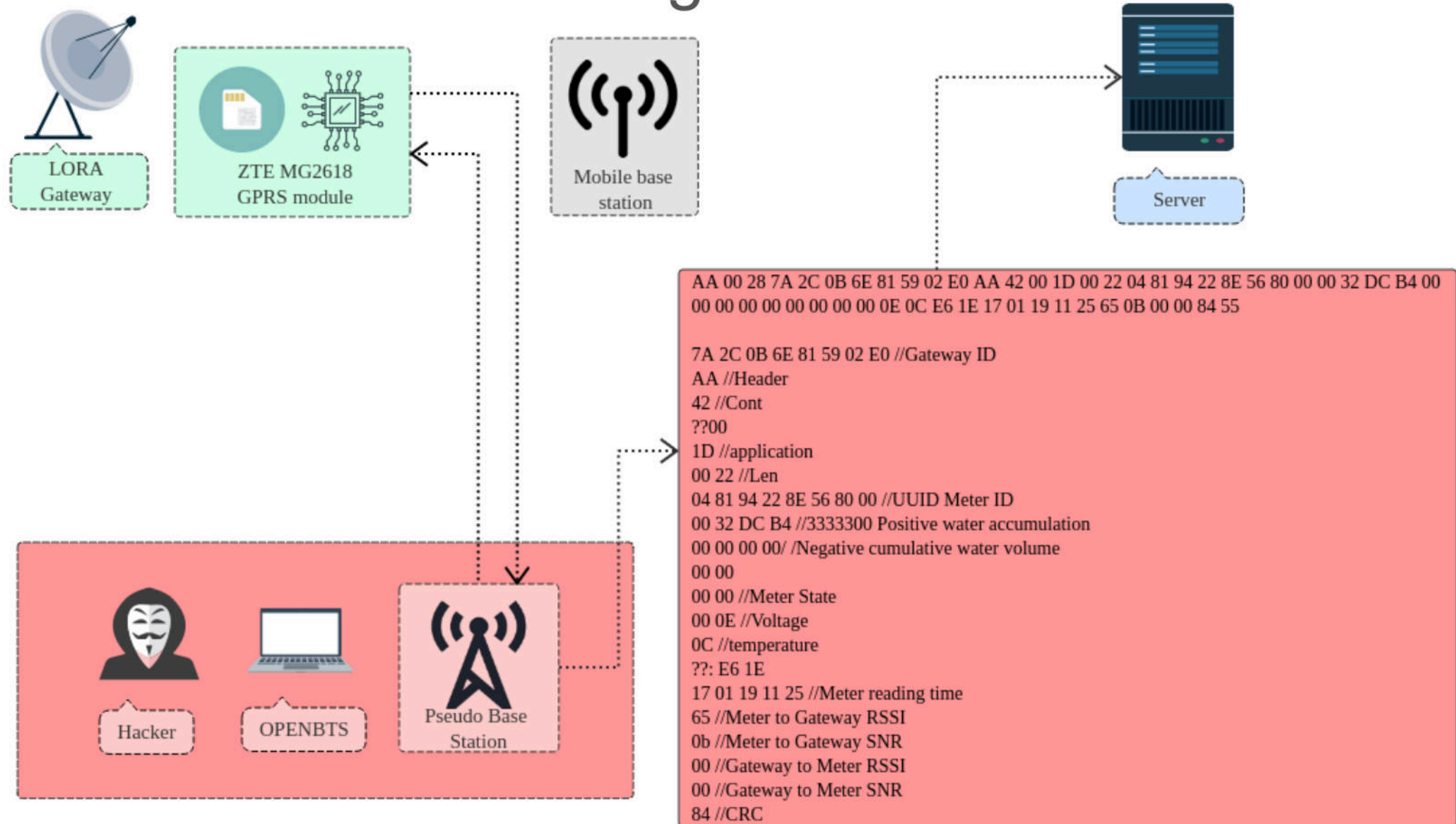
AA	// Header
00 0C	// Len
7A 2C 0B 6E 81 59 0 E0	// Gateway ID
AA	// Header
41	// Cont Code
00	//
02	// Application
00 06	// Data length
6C 78 08 21 01 96	// Data (mcu version:6C 78 radio mcu:08 21 NID:01 CH:96(150))
7E	// Check code =
(0c+7a+2c+0b+6e+81+59+02+e0+aa+41+02+06+6c+78+08+21+01+96)	
55	// END

Reverse Meter upload protocol

```
AA 00 28 7A 2C 0B 6E 81 59 02 E0 AA 42 00 1D 00 22 04 81 94 22 8E 56 80 00 00 32 DC B4 00 00 00 00 00 00 00 00  
0E 0C E6 1E 17 01 19 11 25 65 0B 00 00 84 55
```

7A 2C 0B 6E 81 59 02 E0	//Gateway ID
AA	//Header
42	//Cont
??00	
1D	//application
00 22	//Len
04 81 94 22 8E 56 80 00	//UUID Meter ID
00 32 DC B4	//3333300 Positive water accumulation
00 00 00 00	//00 00
00 00	//Meter State
00 0E	//Voltage
0C	//temperature
E6 1E	
17 01 19 11 25	//Meter reading time
65	//Meter to Gateway RSSI
0b	//Meter to Gateway SNR
00	//Gateway to Meter RSSI
00	//Gateway to Meter SNR
84	//CRC

GPRS module data sniffing



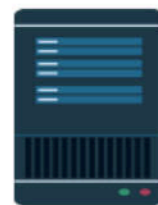
Communications link



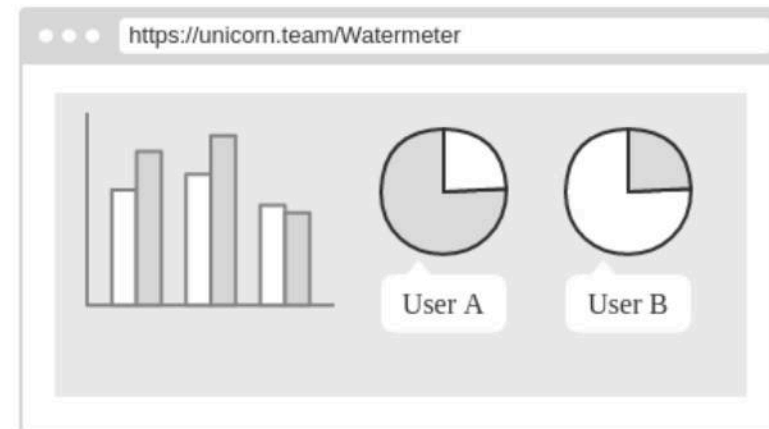
LORA Meter



LORA Gateway



Server

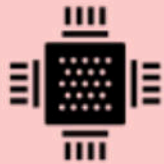
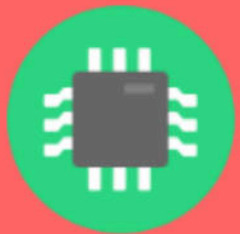


```
0x91 0xF7 0xC0 0x40 0x80 0x18 0x2C 0x81 0x29 0x01 0x00
0x04 0x81 0x94 0x22 0x8E 0x56 0x80 0x00 //UUID Meter ID
0x08 0x00 0x01 0x06
0x29 0x0A 0x00 0x00
0xB4 0xDC 0x32 0x00 //3333300 Positive water accumulation
0x00 0x00 0x00 0x00
0x00 0x00 0x00 0x00
0xE6
0x0C //temperature
0x1E
0x0E //voltage
0x00
0x00 //Gateway to Meter RSSI
0x00 //Gateway to Meter SNR
0xBF 0x38 //END
```

```
AA 00 28 7A 2C 0B 6E 81 59 02 E0 AA 42 00 1D 00 22 04 81 94 22 8E 56 80 00 00 32
DC B4 00 00 00 00 00 00 00 00 00 00 0E 0C E6 1E 17 01 19 11 25 65 0B 00 00 84 55
```

```
7A 2C 0B 6E 81 59 02 E0 //Gateway ID
AA //Header
42 //Cont
??00
1D //application
00 22 //Len
04 81 94 22 8E 56 80 00 //UUID Meter ID
00 32 DC B4 //3333300 Positive water accumulation
00 00 00 00 //Negative cumulative water volume
00 00
00 00 //Meter State
00 0E //Voltage
0C //temperature
?: E6 1E
17 01 19 11 25 //Meter reading time
65 //Meter to Gateway RSSI
0b //Meter to Gateway SNR
00 //Gateway to Meter RSSI
00 //Gateway to Meter SNR
84 //CRC
```


test environment



SX1278



Arduino



Hacker

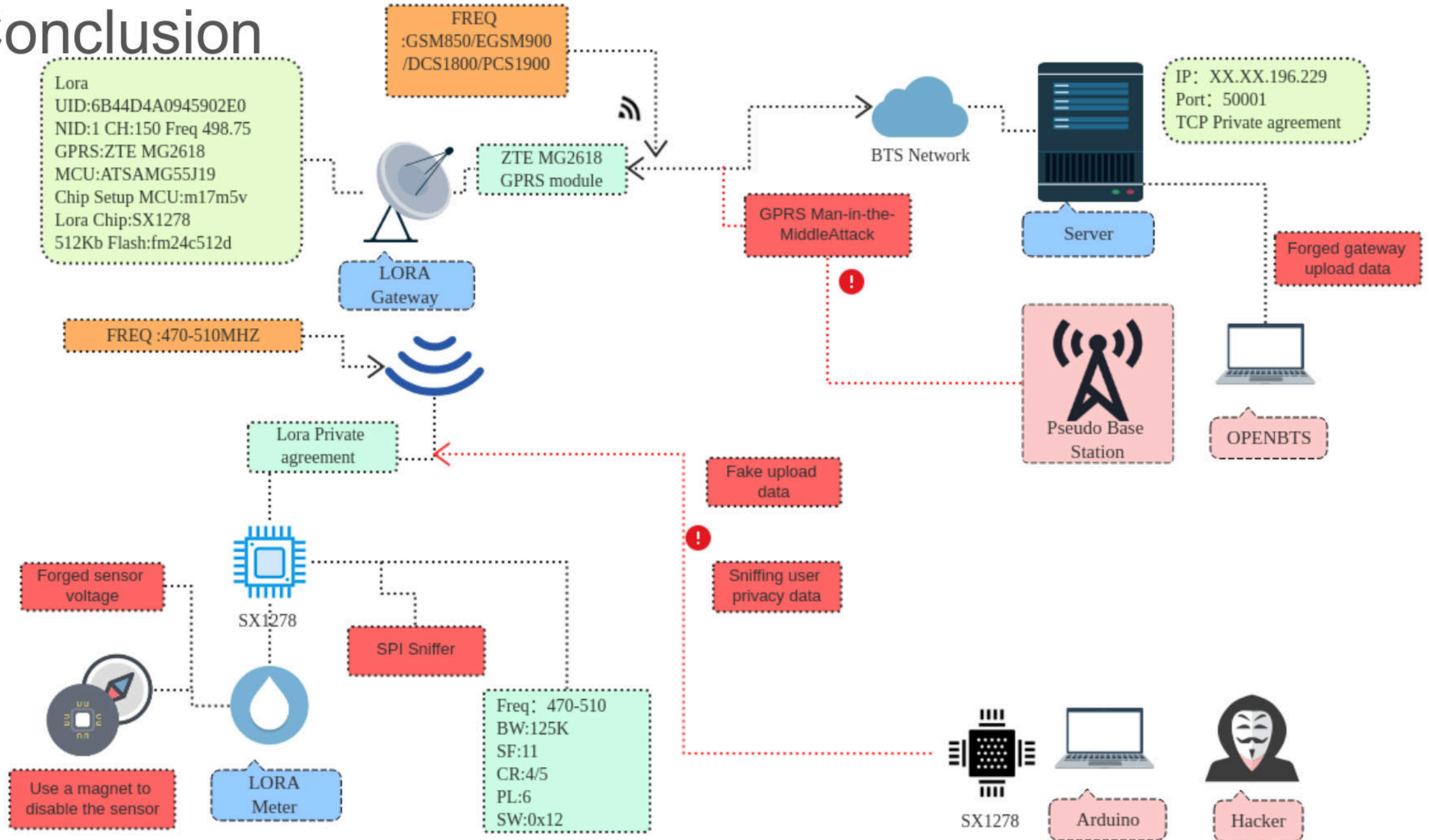


OPENBTS

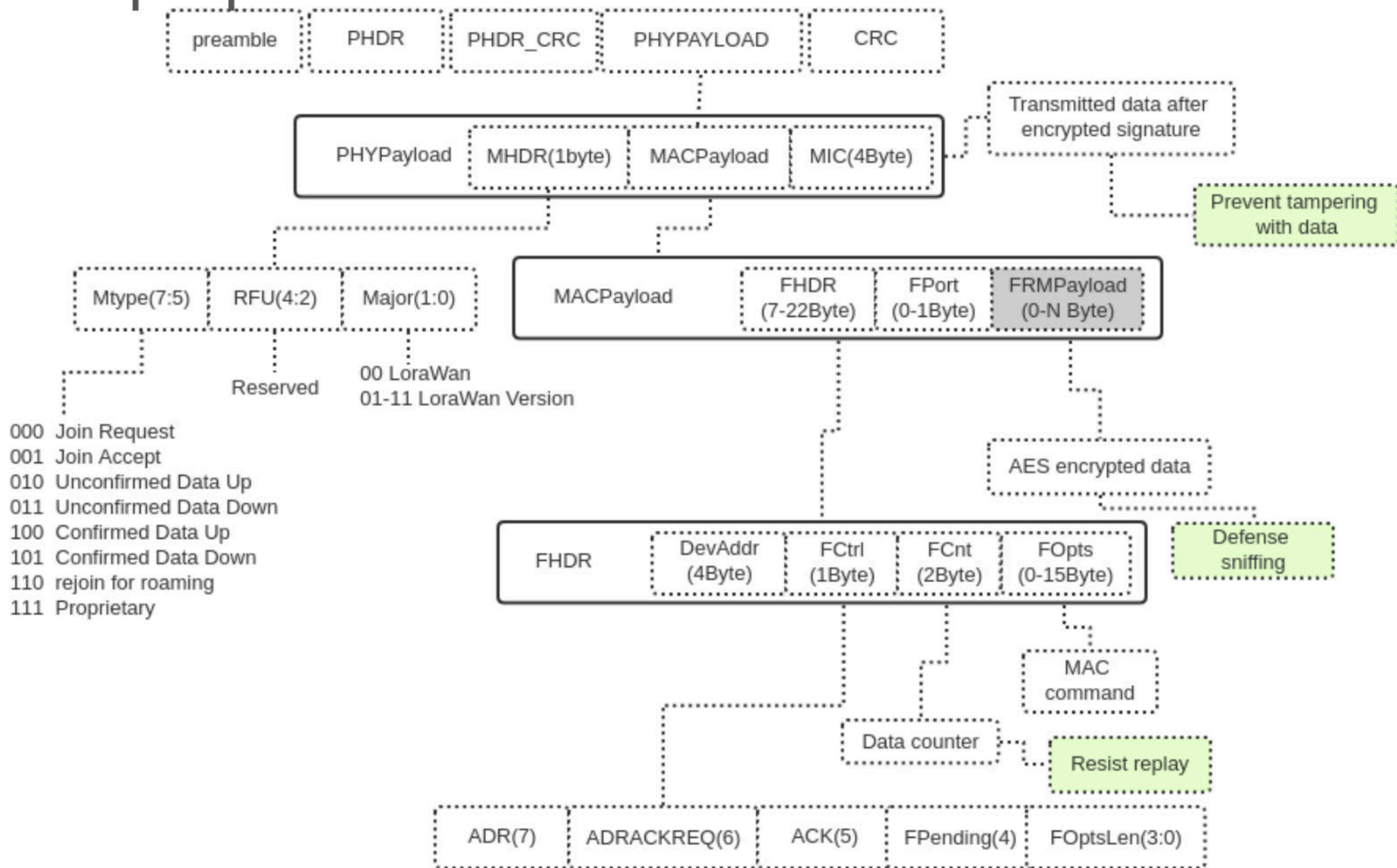


Pseudo Base
Station

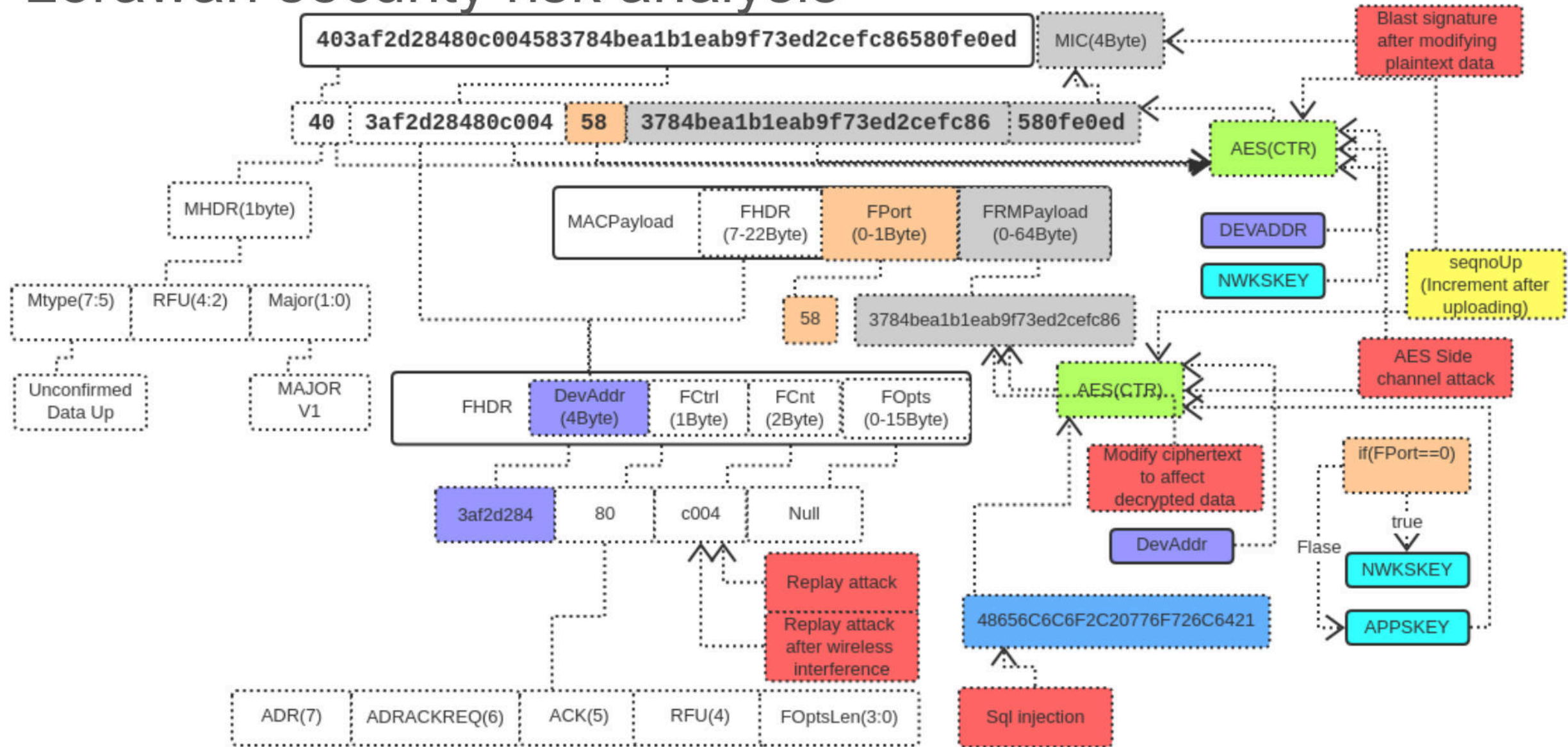
Conclusion



Defense proposal



Lorawan security risk analysis



References

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Thank You!